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Project Repo URL :https://github.com/MarciaSatie/raspberrypi_project/tree/main

Grade Band	IoT Solution	Networking/IoT Technologies	Combined Knowledge	Communication
Base	Use of SenseHat and Camera	(Basic one-way) Use of WIFI to connect raspberry PI, devices, computer and Cloud	Programming Strand: Python scripts. Computer Systems: Pi hardware.	Basic README and project video.
Good	Data processing: Converts sensor data to JSON, uploads images to Cloudinary and streaming to Cloudflare	Interconnected processes: Pi communicates with Flask via MQTT.	Combines Web Development (Flask) with Hardware (Sensors).	Use of Github Repository
Excellent	IoT Application: Prototypical "Smart Security Camera" where you can watch raspberry pi Camera anywhere with internet connection.	Lightweight messaging: Uses MQTT. Network programming: Cloudflare tunnels to bridge local/public networks.	2 Strands: Programming, Networking, and Database (JSON Flat-file).	Learning resources, installation guide, and instructional video in README.
Outstanding	This solution is applicable to real-world applicability for home security.	Cloud Platforms: Cloudinary, CloudFlare,HiveMQ, and Render.	Self-acquired knowledge: Implementing Cloudflare Tunnels and Video Streaming.	Accessible Platform: Live website hosted on Render adn Blynk

Additional Comments:

Grade Spectrum

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Release 1 Base (30-49)	Basic solution that may form basis of overall application. Sensor/device focused.	Physical/Data link layer solution. Basic <u>one way</u> connection between device/processes.	2 <u>programme strands</u> present in output. Basic knowledge of each exhibited. (e.g. programming, database, computer systems)	Zip file and/or basic Repo: Minimal (1) communication resource used (e.g. simple readme.md) and video.
Release 2 Good (50-64)	Solution with clear IoT and domain application. Includes some data processing and/or gateway function.	Wireless/Wired protocols including network and transport layer. >1 protocol. Interconnected device(s) and or processes.	Apply and combine concepts from more than two modules/strands..	Good <u>Github</u> Rep: Repository includes clear structure, documentation.
Release 3 Excellent (65-80)	IoT Application of good prototypical standard. Used to evaluate overall suitability for a full project.	Lightweight messaging. Network/API programming. Architecture/ IOT Framework that mediates between devices and components.	>2 strands as above including more advanced knowledge and concepts.	Excellent <u>Github</u> Repo: Additional communication resources (e.g. instruction video, learning resources, installation guide)
Release 4 Outstanding (80-100)	Novel solution of clear applicability to specific domain. Could result in employment/internship offer.	All <u>previous to excellent</u> level. Excellent Use of Cloud/IoT specific platforms	All above, including self-acquired knowledge over and above module content.	All the above to excellent level, accessible project platform (e.g. web site, Social Media)